Imperceptible Interfaces: Embedding Responsive Systems Through Miniaturization, At Scale.

by

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Contents

Abstract

Current human-computer and human-robot interfaces remain physically separate from the dynamic or soft surfaces that shape our everyday environments. In living organisms, interaction modalities are not attached on top of the skin. Nerves and receptors are woven into skin, muscles and even bones, where they sense and process signals to act locally. Despite decades of ubiquitous-computing progress, most interactive systems still depend on external rigid electronics and bulky connections that, in human and robotic wearables, restrict movement, reduce comfort, complicate mechanical structures, and make technology conspicuous. This dissertation proposes "Imperceptible Interfaces", a framework for embedding sensing, computation, and actuation directly into matter through miniaturization and scalable manufacturing, to weave technology into the fabric of everyday life, at scale.

From biological to synthetic primitives?
Distributed interfaces need a primitive or functional unit. We could call them taxels (tactile elements), or interaxels for general interaction pixels (tactile, acoustic, visual, kinetic, etc): distributed inside the responsive matter, potentially woven into a neuron-like mesh. Aligned with Responsive Environments' research focus on sensing, my prior work built sensate skins in several ways: 3d positioning (HiveTracker - UbiComp'19), in-situ polymerization of textiles (PolySense - CHI'20), sensitive mechanical metamaterials (MetaSense - UIST'21), flex sensor matrices in robotic fingers (RoboTouch - ICRA'23), and even interactive plants instrumented for signal sensing and electro-stimulation, giving the plant a cyborg nervous system (Pudica - AHs'22). However, biology can inspire us for further actuation frameworks. For example, Cicadas click by buckling a ribbed membrane in their body wall. Synthetically, embroidered speakers can follow this logic and produce audio from the fabric itself, without magnets (SonoFlex - UIST'20). Also, the cuttlefish changes appearance through motor neurons and pigment sacs. My explorations on thermal LCE actuators (FibeRobo - UIST'23) and my photochromic textures (PortaChrome - UIST'24) echo this synthetically.

From sensate skins to nervous and muscle fibers?
This work led to my core thesis project (FiberCircuits, UIST'25), a miniaturization framework for flexible, fiber-like circuits integrating processing, sensing, and feedback modalities. Rather than attaching electronics onto surfaces, FiberCircuits weaves them within the structural matrix. The fiber is a particularly constraining geometry, but also a key that opens many locks: what fits inside a thread will fit inside a skin, a muscle, a seam, or an endoscopic gripper. Today's version uses millimeter-scale parts on narrow flex circuits, encapsulated, braided, and integrated through embroidery, weaving, or knitting. The proposed research extends it to a sub-millimeter system-in-fiber model with silicon bare dies, electrodes, encapsulation, etc. The main investigation is high-density tactile I/O: dense taxel arrays combining touch sensing, processing, feedback and communication along thin fibers. These sensate "nerves"

could eventually control synthetic muscles as a longer-term exploration: a horizon where sensing, computation, and actuation are all synthetic and integrated into the same structure. In human and robotic wearables, localized nodes improve latency and power distribution, and can form sensitive computational networks at the material scale across garments, robotic skins, or fiber muscles. As realizing this vision is beyond a single dissertation, my aim is to establish the system architecture and manufacturing methods for many more scientific endeavors.

Re-purposed manufacturing for deployment and impact? In functional fiber research, the classic approach is to draw macroscopic preforms into microscales. This process is delicate, density-limited, and reliant on specialized equipment that few facilities have. FiberCircuits takes a different route: pushing the existing flex circuit manufacturing process, initially built for consumer electronics, which lowers cost and enables researchers, artists, and small companies to reproduce it. A key objective is to move beyond packaged components towards processor bare dies assembled onto high-density flex substrates via flip-chip bonding, investigating wafer-level access, die placement, interconnect geometry, encapsulation, and continuous fabrication at volume. An important goal is to use existing manufacturing ecosystems and (re-purpose machines to) enable scalable reproduction by others from academia to industry. The initial explorations are supported by collaborations with the MIT Lincoln Laboratory for flip-chip evaluations, and amplified with industry research labs at Amazon and Meta, for applications in embedded interaction and robotics.

Contributions, dissemination, and impact, from the lab to applications. Two evaluation directions are planned. Firstly, manufacturing: yield, electrical performance, durability, washability, cost, throughput. Secondly, interaction: human-subject studies of comfort, biosignal quality, and eyes-free interaction intuitivity, which would be tested in real settings such as instrumented sleep masks, and robotic manipulation. This dissertation proposes to explore research questions around this core investigation: How far can synthetic nerves be miniaturized for human and robot interfaces to achieve imperceptibility? Expected contributions include a scalable manufacturing framework for miniaturized e-fibers, design guidelines for system-in-fiber architectures, demonstrations across human and robotic wearables, plus open source designs, documentation, and evaluation methods. Beyond research outputs, these explorations are disseminated through yearly workshops at CHI, UIST and UbiComp, about Open Hardware and Intelligent Soft Wearables. This scaling philosophy drives me to push the impact further by running yearly symposia and residencies, to help researchers and entrepreneurs scale their work. These programs reach hundreds of researchers across academia, industry, and art, also equipping dozens of MIT participants with methods they carry into their own labs and career. Together they push responsive systems from lab prototypes toward robust, scalable use, in the real world.

1 Problem Statement

1.1 Context

1.1.1 Two Bodies, One Problem: The Integration Gap

My grandparents had fall detectors with alert buttons and they almost never wore them. The pendants were bulky, and they announced to the room that someone was old and fragile, so they stayed in a drawer. The best sensor in the world is worthless in a drawer. They are not exceptions. Falls are a leading cause of injury in older adults. Reliable detection is getting accessible on smartwatches, but not everyone loves wearing them. Similarly, that device did not fail because its electronics were bad, it failed because it was rigid, visible, and uncomfortable, so it was abandoned.

Our own bodies show the alternative. We do not wear our sense of touch. Nerves and receptors are woven into the skin, where they sense, process, and act on the spot. Nothing is bolted on, nothing has to be remembered in the morning. The interface disappears because it is part of the entity it serves.

The same problem appears on the synthetic body. A robotic hand can be covered in sensors, but each one usually needs wires back to a controller. Long cables lose power, degrade data transmissions, and damaged connections can kill a finger. Here too the fix is the biological one: put the sensing and processing where the work happens, and distribute it, turning a wired sensor array into a distributed sensate medium[1].

The human and the robot are two versions of one problem. In both, we still lay technology over a soft, moving body instead of embedding it. In both, the costs include reduced comfort, robustness, and usability, ultimately contributing to abandonment. This dissertation proposes to close that gap under the framework of Imperceptible Interfaces: implanting sensing, computation, and feedback directly into the substrate, through miniaturization and scalable manufacturing, for human and robotic bodies alike. One such functional unit is what may be called an in-situ *Interaxel*, a small interaction cell embedded throughout the body.

1.1.2 The Interface That Would Not Disappear: A Retrospective

Mark Weiser famously envisioned technologies that would *disappear and weave themselves into the fabric of everyday life* [2]. Decades later, they have not disappeared yet, leaving the vision a rich and open opportunity.

Long before this vision, Claude Shannon and Edward Thorp built in the 60's what many consider the first wearable computer [3]: a cigarette-pack-sized device (see Figure 1.1-left),

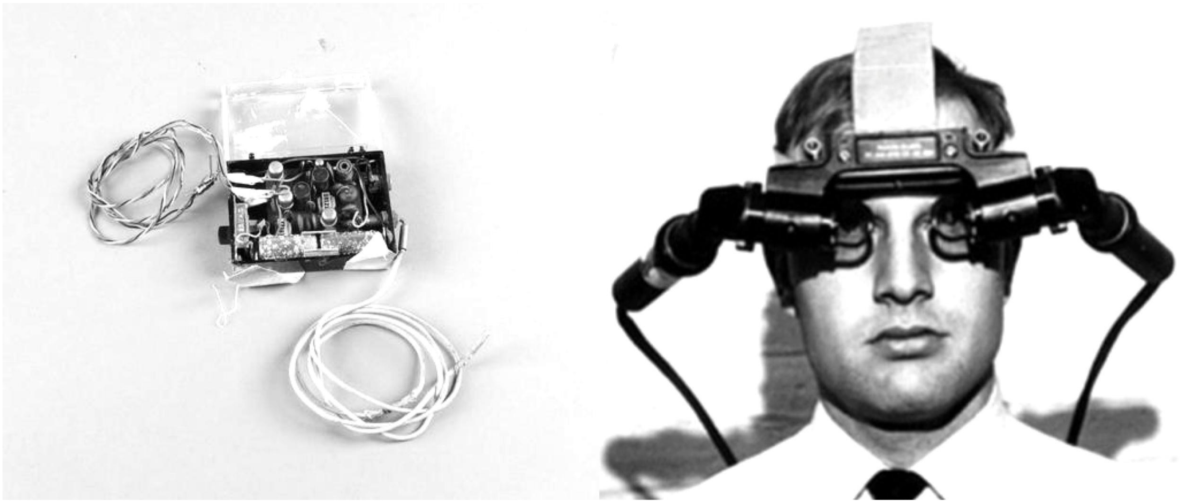

Figure 1.1: Left: the first wearable computer (1961) [3] - Right: the first head-mounted display (1966) [4]

concealed on the body to help play casino roulette. In the same decade, Ivan Sutherland built the first head-mounted display [4]. It hung from the ceiling on a mechanical arm, so heavy it was nicknamed the Sword of Damocles (see Figure 1.1-right). Both were visions of computing worn on the body. Both were also rigid interfaces strapped onto the user.

Thirty years later, the interfaces had shrunk but not softened much. The wearable-computing groups of the 1990s carried real computers on belts and backpacks, with displays over the eyes and keyboards in one hand (see Figure 1.2-right). The vision was radical and the engineering was meticulous, but the result still sat on top of the skin, very visible.

This thesis work takes Weiser's words literally, and it is not the first at the Media Lab to do so. In 1997, Post and Orth introduced Smart Fabric [5], building circuits from conductive textiles for data, power and touch sensing, arguing that hard plastic cases against the body were the wrong path. This work grew into e-broidery [6], which suggested the use of round chip package, meant to be sewn into fabric and mass-produced (see embroidered jacket in Figure 1.2-left). That ambition, moving past packaged parts toward bare silicon in a soft, manufacturable substrate, is the thread this thesis carries, down to a sub-millimeter scale. Other groups pursued e-textiles research in parallel, for example the knitted physiological-sensing garments [7], though often relying on better-resourced facilities.

Meanwhile, more radical explorations were pushing the miniaturization to its physical limit. In 1999, Kris Pister proposed Smart Dust: a complete sensor, power supply, processor, and communication link integrated into a single cubic millimeter [8], [9]. The speculation was that computing could become so small it would scatter into the environment like dust - sensing everything, seen by no one. Smart Dust has been a bet on miniaturization as the path to invisibility, and much of it came true in sensor networks.

Smart Dust proved how small computing could become. This thesis builds on that promise and asks where those grains could live: not scattered loose, but held conforming to organic or synthetic bodies. The opportunity is to push the miniaturization further and add structural integration, giving each tiny node a place such as in a robot glove, or underwear. This thesis works toward that step, implanting and manufacturing sensing into conformable structures, at dense scales, low cost, and reproducible by others.

Figure 1.2: Mid 90's Wearable Computing researchers at MIT, including Rehmi Post (extreme left) wearing the first wearable e-Textile [5], and others wearing body-mounted computers / displays, one-handed input devices, etc.

1.2 Background: From Sensate Skins To Nervous Fibers

The previous section argued for sensing that conforms to the body. This one narrows the focus: addressing what the field has recently built toward that goal, and why does each approach stall before reaching imperceptibility. Prior work sorts into three families. Each solved part of the problem in a different way, and each hit a different wall. The foundation of this thesis, FiberCircuits [10], was designed to sit where the three meet.

1.2.1 Functional Materials For Interactive Textiles

The first route makes the fabric itself the sensor. As introduced above, early work at MIT demonstrated capacitive sensing directly through embroidered conductive textiles [5], [6]. It later grew into expressive materials and custom piezoresistive matrices [11], [12]. Thread-level work pushed sensing from one dimension [13] to double-sided layouts [14] and mid-air fields [15]. Finally, functionalization by in-situ polymerization (PolySense [16]) added new sensing capabilities to textiles. This approach nonetheless hits a wall: textile-only solutions cannot perform advanced functions such as inertial or magnetic sensing, and they cannot do local processing. Without computation in the material, every signal must run out to an external rigid board.

1.2.2 Integrating Circuits Onto Textile

The second route adds real silicon. Rigid PCBs bring full processing but sit on the surface, obtrusive and uncomfortable, so designers have to explore alternatives to conform to the body [17]. Flexible PCBs improve draping and enable distributed sensing with processing (Chainmail [18], SensorTape [19], SkinKit [20], SkinLink [21]). A related idea moved past the flat board and embeds sensing into the load-bearing structure itself: MetaSense integrates sensing directly into 3D-printed mechanical metamaterials [22], so the object's geometry becomes the interface. The wall: most of these remain surface-mounted and prone to snagging, and bulky interconnects limit conformability, which is eventually tied to user abandonment [23], [24].

1.2.3 Advanced Fibers

A third route embeds electronics into the fiber. On the materials science side, breakthroughs gave single fibers piezoelectric or electroluminescent behavior [25], [26]. Komolafe et al. integrated discrete components onto thin circuits encapsulated into yarns [27], an early proof that packaged parts could live inside a thread, though with limited capabilities. Loke et al. drew digital logic into fibers [28], but their thermal draw tower is inaccessible to most labs or manufacturers, hard to customize, and sparse in density. For density, Hwang et al. used a semiconductor fab process to integrate components on a microfiber [29], but few researchers or manufacturers can replicate this approach. Fibers can actuate as well as sense: drawn liquid-crystal elastomer fibers [30] and fluidic fiber muscles [31], [32] show that a single fiber can move; an actuation direction that this thesis touches, but keeping it peripheral. The wall: these depend on thermal-draw towers, semiconductor facilities, or bespoke machines for each new function. That tooling is costly and limited in versatility and yield, with poor manufacturing accessibility.

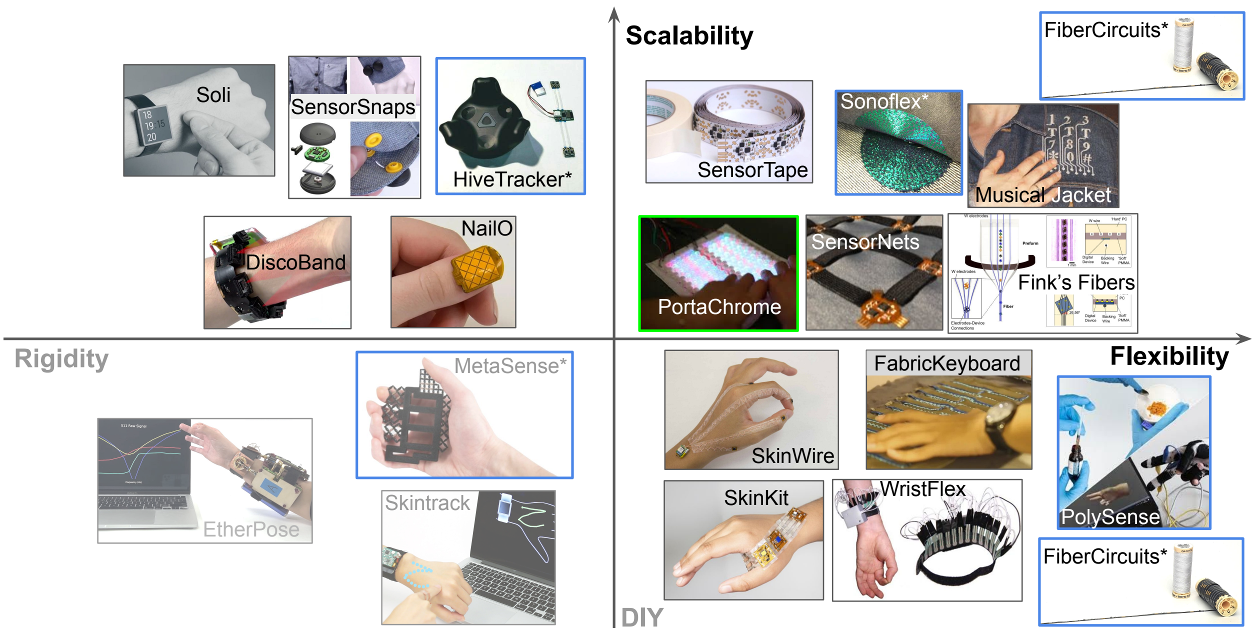

Figure 1.3: Comparison of selected prior work and personal contributions (marked with a star*)

1.2.4 Closing the Gap: FiberCircuits as the Thesis Foundation

Flexibility being a key challenge for conformability and acceptance, scalability also matters for impact. This thesis is built on work that explored various approaches (see examples in Figure 1.3) and grew into the foundation that this thesis extends: FiberCircuits [10] addresses the gap by miniaturizing conventional flexible PCBs into yarn-scale strips, coated and braided, so production leans on globally available fabrication techniques. This thesis builds on that, pushing from miniature parts originally for high density electronics such as those found in smartphones. However, a similar wall appears for robot skins, conforming poorly and adding failure points, so distributed sensing in the material serves both bodies, and this proposal treats them as one problem.

1.3 Solutions and Opportunities: From System-on-Chip to System-in-Fiber

The aforementioned gaps point to a clear target: embedded processing and sensing that is dense, conformable, and manufacturable by others.

This section builds the response in four steps. It first generalizes a primitive from personal contributions, the interaxel. It then argues that discretizing these primitives for conformability can be achieved through miniaturization. It next examines why a fiber is a particularly advantageous geometry, introducing FiberCircuits as a first exploration framework. Finally, it formalizes the approach as a paradigm shift from System-on-Chip to System-in-Fiber.

1.3.1 Interaxels: Personal Contributions & Primitive Generalization

A system primitive can be generalized for in-situ interaction elements, as observed in personal contributions that led to this thesis (Figure 1.4). On the sensing side, HiveTracker [33] miniaturized 3D positioning; PolySense [16] functionalizes textiles for touch, stretch, temperature, and humidity sensing; MetaSense [22] senses structural deformations; RoboTouch [34] was a first robot skin to sense shapes; FiberCircuits [10] senses touch, orientation, and motion; finally, DexteriSense (in progress) targets high-density sensing for robot skins.

On the feedback side, SonoFlex [35] turns fabrics into speakers; MagKnitic [36] adds knitted haptics; PortaChrome [37] borrows the chromatophore for color; Ferrozuit [38] uses magnetic fibers for zero-gravity suits; Tactile-IO (in progress) proposes to explore miniature touch sensing and electro-stimulation.

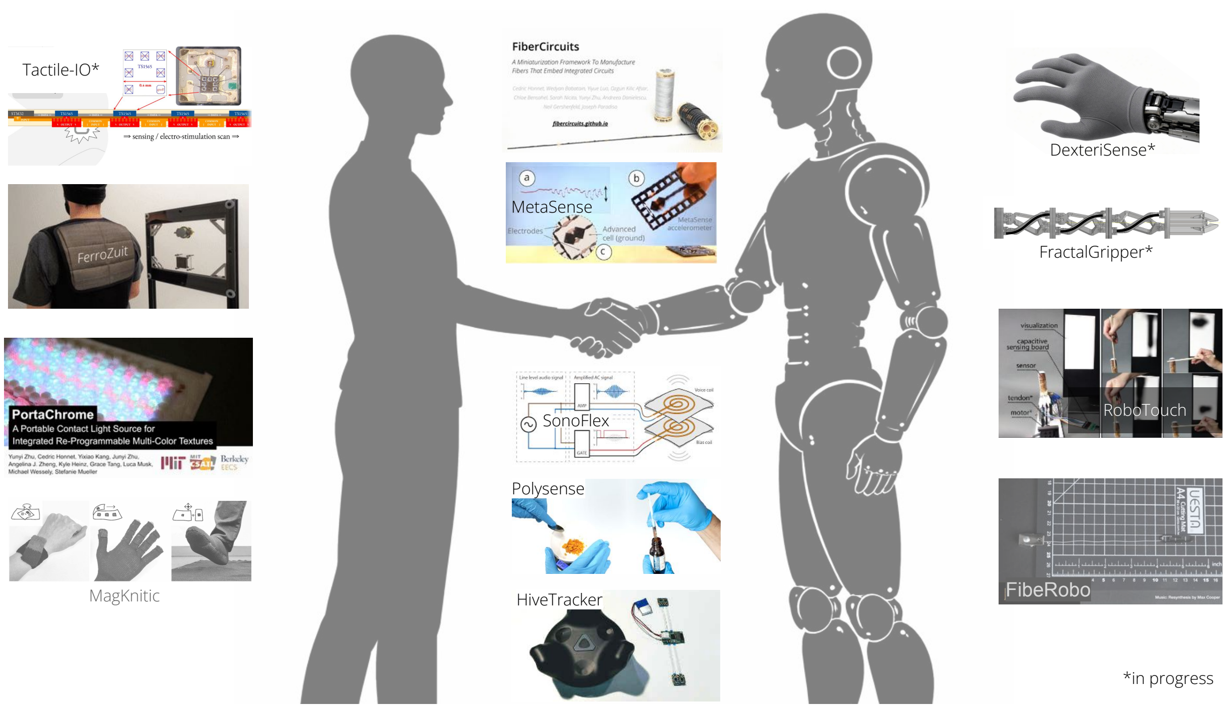

Figure 1.4: Personal contributions and primitive generalization from examples of situated interaction elements, for humans, robots or both.

Across the different contributions listed above, the same pattern recurs: an element that senses, processes, and/or responds, placed *in-situ* within the body structures it serves. The most classic interactive elements are the familiar pixels (picture elements), but concepts such as taxels (tactile) and haxels (haptic) have also been used in HCI. This thesis proposes to generalize the idea with the *interaxel*.

1.3.2 Miniaturization and Discretization to Enable Conformability

Interactive elements can be bulky and rigid. From processing, sensing, or feedback modalities, bulky rigid blocks cannot follow curved or morphing bodies. Discretization splits monolithic functions into multiple small nodes with similar overall functionalities (Figure 1.5), so the aggregate can drape over a joint or a fingertip that no single part could follow.

For example, bulky processors or memories can be split and serialized with multiple smaller ones. Same goes with IMUs that integrate an accelerometer and a magnetometer for example. Such IMUs do not have off-the-shelf options below 2 mm wide (example: LSM303AH), but composing the functions from separate parts can enable serialized IMU-like systems that are sub-millimeter wide (0.8 mm examples: BMA530 and iST8306). Distributing one component across a fiber introduces registration drift, which can be corrected with differential referencing [39].

Miniaturization and discretization can also enable new modalities for high density sensing, which can be useful in magneto-micrometry [40] for example. Together these discretized node assemblies turn stiff functions into conformable ones.

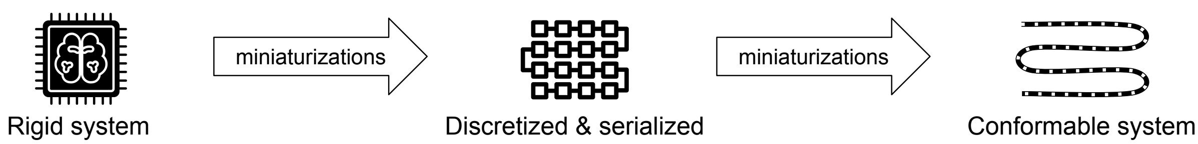

Figure 1.5: How conformability can be enabled by discretization and miniaturization: a rigid interaction element can be split into multiple miniature nodes with the same overall function.

1.3.3 Why a Fiber: FiberCircuits as First Exploration Framework

While assembling interaxels for responsive skins, the flexible surface choice might feel obvious, but a fiber can actually be more advantageous:

- **Conformability.** A thread can bend in more directions than a flexible surface so it can closely follow a morphing body.
- **Topology.** Fibers can also adopt special topologies in complex structures such as sensing gloves, for humans or robots.
- **Insertability.** Like a nerve, a fiber can thread into places a flexible surface cannot: a seam, a glove finger, a narrow channel, etc.
- **Chainability.** One long fiber can be cut to length or chained, so the same linear module becomes a wristband, a longer seam, or a matrix.

Smart fibers also share advantages with responsive surfaces that might be less obvious:

- **Parallelization.** Perpendicular fiber connections, for power or data distribution, can avoid voltage drops or synchronization problems along long PCB traces, as seen in body-scale sensor networks research [41].

- **Scalability.** A one-dimensional substrate can use existing roll-to-roll or textile manufacturing ecosystems, rather than having to find special tooling.

FiberCircuits is a first exploration for the framework of this thesis: interaxel arrays on flex PCBs that are thin enough to be woven or knitted into ordinary textiles. Figure 1.6 illustrates the idea, but the publication [10] addresses details such as coating and washability, mechanical robustness evaluation, scalability with continuous production possibility (roll-to-roll), and real-life application scenarios. However, there is an even higher potential for advanced miniaturization using bare dies of integrated circuits, to enable high-density interaxel systems, for example with human or robot wearables. This is discussed in the Research Plan below.

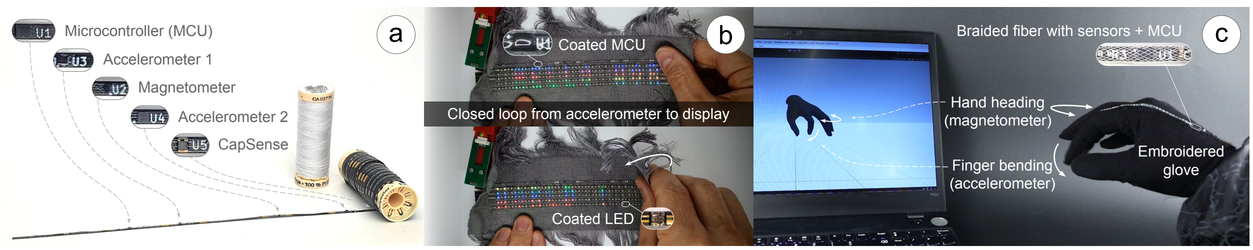

Figure 1.6: (a) The main FiberCircuit illustrated as a sewing thread with embedded processing and sensing (two accelerometers, a magnetometer, and capacitive sensor). (b) A woven sample with 250 LEDs scrolling "Hello World" at a speed controlled by the accelerometer. (c) A FiberCircuit embroidered on a glove to control a VR hand with touch, hand orientation and finger bending.

1.3.4 A Paradigm Shift: From System-on-Chip to System-in-Fiber

Decades of miniaturization enabled integrating entire computers onto a single die, the *System-on-Chip*, often with 3D stacking. This thesis proposes another direction: extending that die along an effective thread, into a *System-in-Fiber* that distributes computation, inputs/outputs, memory, and power management (Figure 1.7). Distributing inference across such nodes extends frameworks we explored for embedded distributed machine learning [42]. The components stay rigid but they are now miniaturized, the encapsulation stays waterproof but it is flexible, and the assembly becomes long, thin, and conformable. This proposal aims to develop and evaluate this architecture, on both human and robotic bodies.

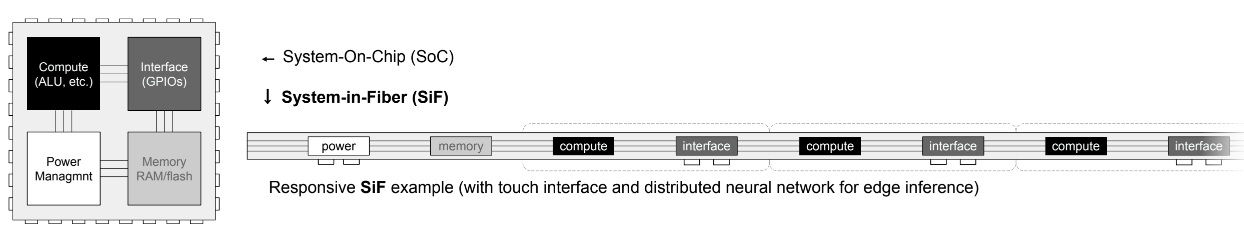

Figure 1.7: From System-on-Chip to System-in-Fiber: the same functions (such as compute, I/O, memory, power management) unfolded from a single die along a one-dimensional substrate.

2 Research Questions

The previous chapter argued for a System-in-Fiber: dense, conformable, manufacturable interaxel fibers. This chapter explores feasibility questions, risks and alternatives. Regardless, the overarching question will remain, for HCI and Robotics applications:

> *How far can we miniaturize synthetic nerves with distributed sensing and feedback, to enable human and robotic wearables that approach imperceptibility?*

RQ1: Manufacturing a body-coupled substrate. Can interaxel fibers be manufactured at high density, coupled reliably to the body, and be replicated with accessible tools?

(a) How small can electrodes be for capsense or electro-stimulation, and what materials do they need (printed electrodes, conductive silicone, micro pogo-pin contacts)?

(b) How much can sub-millimeter bare-die FPCs with flip-chip assembly improve density over drawn fibers, while keeping cost, reliability and reproducibility accessible?

RQ2: Sensing and stimulating along a fiber. Given such a substrate, can a one-dimensional array sense and stimulate richly enough for real interaction?

(a) Can dense sub-millimeter interaxel arrays reach the spatial and temporal resolution needed for eyes-free interaction on the body, what are the driving voltage or shielding implications, and how does this density perform on a robotic hand?

(b) For electro-tactile feedback (haxels), which frequencies and duty cycles are most legible, how much would conductive encapsulation and weaving limit the sensation, and how does perceived intensity compare with a linear resonant actuator at similar power consumption?

RQ3: Imperceptibility in use. Does building the interface into the material, rather than mounting it on the surface, measurably change whether it is usable and kept in use?

(a) Do the interfaces hold up in real settings, such as sleep sensing (eye mask, pillow, or wristband, checked against a reference such as an open smart ring) and robotic manipulation, measured on touch-sensing accuracy?

(b) If people wear a rigid wrist wearable, then a conformable woven one for a given time, is the experience difference enough to argue that conformability lowers abandonment risk?

Together the three questions trace one thread from fabrication to felt experience: RQ1 makes and evaluates the substrate, RQ2 tests how well it senses and stimulates, RQ3 tests how much it translates into imperceptibility, on both human and robot bodies. The research plan in Chapter 3 pursues them through concrete prototypes and evaluations.

3 Research Plan

This chapter lays out how each question will be explored: what will be built, how it will be evaluated, and what the likely risks and alternatives are. It follows the same logic as for the Research Questions in Chapter 2: build the substrate (RQ1), test whether it senses and stimulates (RQ2), then deploy it and study its use (RQ3). As in any research project, parts of it may not work as hoped; each step below lists a fallback, so inconclusive results still yield a useful finding.

The work centers on one prototype family, which we call **Tactile-IO**: sub-millimeter taxels for sensing, and electro-tactile haxels for feedback, extending FiberCircuits [10] from classic packaged parts toward bare dies. This approach is also planned to be explored in an industry context for high-density sensing skin on a robotic hand to study dexterous manipulation.

3.1 RQ1: Building the Substrate

Skin coupling (RQ1a). This first task is to study the electrodes, where miniaturized circuits meet the skin. The plan is to compare electrode variants down to sub-millimeter footprints, across materials and contacts such as printed classic PCB pads, conductive materials and micro pogo-pin contacts, for both sensing and electro-tactile stimulation.

Evaluations: contact impedance, sensing SNR, and stability across repetitions.

Risk: voltage drop may degrade electrostimulation and sub-mm electrodes may be too small for effective sensing, but conductive thread could extend them as a fallback option.

Advanced miniaturization (RQ1b). The second task is to move from classic packaged parts toward 0.4 mm bare-die placement and flip-chip bonding on narrow flexible PCBs, in collaboration with Lincoln Laboratory for local flip-chip process experimentation. A next step would be to implement this process by re-purposing the high precision machines used for stiffener PnP, in collaboration with a manufacturing partner in Shenzhen (iStar). Figure 3.1 shows initial prototypes explored with an advanced manufacturer (SCI Shenzhen) that can do flip-chip assembly and custom packaging, but the costs or minimum orders are generally higher. These process options would allow replication without the need to own advanced machines, and would be compatible with roll-to-roll production for scalability.

Evaluations: interaxel density against the classic packaged-part baseline, fabrication yield, mechanical robustness, and cost comparisons.

Risk: bare-die assembly is demanding, and if density or yield disappoints, an alternative is still in progress using 0.86 mm wide MCUs (MSPM0C1104).

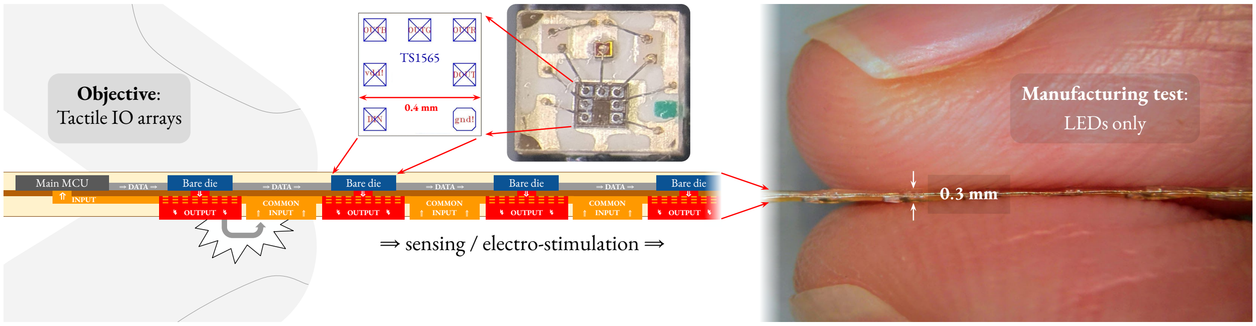

Figure 3.1: Illustration of the first test of the miniaturization potential using a 0.4 mm bare die processor from an addressable LED. This bare die can control LEDs, but also electrodes to enable sensing / electro-stimulation or to activate / deactivate other ICs in a daisy-chain topology, forming dense interactive arrays without length constraint

3.2 RQ2: Sensing and Stimulating Along the Fiber

Input resolution, on both bodies (RQ2a). The goal here is to characterize how finely a one-dimensional taxel array resolves touch in space and time, then to test it in two settings. On the human body, a psychophysical study can measure spatial and temporal thresholds, compared against a non-wearable reference system. Figure 3.2 illustrates potential applications across various sensing and feedback modalities, using the examples of space suits and sleep research interfaces. For robotic bodies, the same substrate would be applied as a sensing skin on a robotic hand and measured during manipulation.

Evaluations: spatial and temporal thresholds, and robotic contact-detection accuracy.

Risk: it is not yet clear how close one-dimensional sensing compares with two-dimensional sensing on fine tasks; measuring that gap is part of the contribution.

Feedback output (RQ2b, higher risk). The aim here is to add electro-tactile feedback, closing the loop so that input and output share the same location for intuitive interaction. A first study is to sweep stimulation frequency and duty cycle, to measure how the sensation depends on skin dryness, and how much encapsulation or weaving attenuates it. It will also be useful to compare perceived intensity and power against a linear resonant actuator, building on prior electro-tactile work [43]–[45]. With sensing and feedback combined, an eyes-free interaction study can then measure how well people locate and act on cues along the fiber without looking.

Evaluations: detection and comfort thresholds, attenuation with dry skin or woven electrodes, perceived intensity vs. output voltage, and eyes-free interaction precision.

Risk: reliable woven electro-tactile output is not yet established at this scale and it may not work, in which case feedback falls back to bigger haxels, while the thesis remains mainly focused on the sensing substrate of RQ1 and RQ2a.

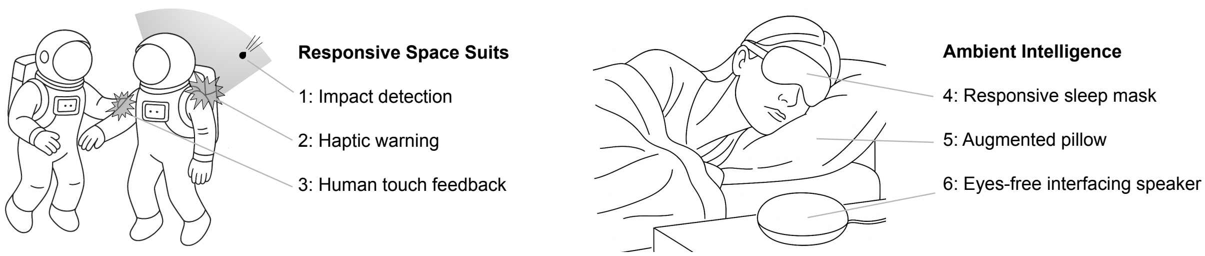

Figure 3.2: Illustrating potential applications from space speculations to sleep research interfaces.

3.3 RQ3: Imperceptibility in Use

Evaluation by real use (RQ3a). The interfaces will be deployed in the recurring application settings of this proposal. For sleep sensing, an instrumented eyemask, pillow or wristband would combine taxels for temperature, capacitive sensing for position, and IMU for movement, with electro-tactile output where RQ2b permits. Sleep stages would be estimated and checked against a reference such as an open-data smart ring, so the fiber-sensor estimates can be compared with a reference signal. If validated, the robotic skin from RQ2a would run on manipulation tasks and report on contact-detection reliability.

Evaluations: agreement with the sleep sensing reference signal, robustness across real wear, and robotic contact reliability tests.

Risk: performance may drop away from the bench, and where it does, the gap between lab and field is itself the finding, with settings simplified to isolate the cause.

Adoption and abandonment (RQ3b). The plan is to run a user study comparing a conventional rigid tracker or smartwatch against a conformable woven one, each worn for a matching period. It stays scoped to comfort, preference, and continued wear rather than clinical efficacy, drawing on documented adoption barriers for body-worn devices [23], [24].

Evaluations: reported comfort and preference from a dozen users, together with self-reported and measured wear durations.

Risk: users may not particularly prefer the woven version, or have specific preferences for unexpected reasons, and either outcome is informative about the imperceptibility claim.

3.4 Evaluation Philosophy

Evidence will be gathered on multiple levels, so that no single result carries the whole argument. Process metrics such as density, yield, cost, durability, washability would allow evaluating manufacturability. Use metrics such as deployment robustness and adoption would test how imperceptible these interfaces made with these fiber substrates can be. An important objective is not so much to prove that they are always better, but to learn where they work best and where they are limited.

4 Conclusion

Why Imperceptibility Matters

As discussed, Weiser defined profound technologies as those that weave themselves into daily life until becoming indistinguishable from it [2]. He was a philosopher before becoming a computer scientist, and disappearance for him was not only a matter of size, but of attention. This is what phenomenology calls the lived body, it separates the body as a physical object from the body as we experience it from within, the one we act through without attention. For example, a cane or a bicycle, once mastered, is no longer felt as a tool but becomes an extension of the body. That is an ideal standard for "imperceptible". In the fall-detector wearables examples, the adoption failure was not because their electronics were poor, but because they never crossed from object to lived body. This reframes the technical work as a means to a perceptual end. Sub-millimeter taxels, body-coupled electrodes, and one-dimensional interaction are explored here because devices that follow the skin and ask for no visual attention have a chance of being felt like parts of the body, not devices on it.

From Nerves to Muscles

The proposed System-in-Fiber architecture also suggests directions that this dissertation will not develop. If each interaxel carries local processing, dense arrays could process signals in place and communicate with neighbors, forming a computational mesh through a garment or robotic skin, closer to a nervous system than a central bus. Actuation is a natural companion that was also explored during this research: drawn liquid-crystal elastomer fibers [30] and next versions of fluidic fiber muscles [32] for robotic grippers show that a single fiber can move as well as sense, and these nerves will soon drive such muscles. These explorations are left as secondary work, the sensing substrate being the necessary first step.

Dissemination, Scalability, and Impact

Scientific impact can only grow better if our research communities have deeper opportunities to exchange actively and build on each other's work, which is why we organize the yearly *Intelligent Soft Wearables* workshops at UIST and UbiComp/ISWC [46]–[48]. There is a deeper problem, though. We found that open hardware is often not sufficient for replication, and our impact could be improved further: reproducing research needs real understanding of *how* it is made. This motivates our CHI workshop on open hardware [49]. Affordability is also part of the imperceptibility argument: a device can only *disappear* better if it can be forgotten as easily as a bandage or a hair tie.

Figure 4.1: Scaling impact beyond individual outputs: symposiums and research residencies about manufacturing, to help others make their work reproducible, affordable, and impactful.

Amplifying the manufacturability of our research can enable that affordability, and can even help to spread our work better: researchers, artists, and entrepreneurs can reproduce it wherever they are, even without fabrication tools. My research led me to spend considerable time in these manufacturing ecosystems, and it enabled me to start organizing manufacturing symposiums and research residencies[1] that teach not only how to build at scale, but also enable us to discover a hugely underestimated "unknown unknown". Hundreds of people benefit from these programs every year (Figure 4.1). Beyond publications that were enabled, we even saw companies[2] sprout in our program, and return as yearly sponsors.

Weaving Hybrid Threads Together

This work sits where art and science meet. Music was the obsession that started everything: it led me to study engineering, and building wearable controllers, which led me to textiles, a lineage that began at the Media Lab in the late 90's [5]. This research also produced art, not just papers: from installations at Ars Electronica about textiles[3] and music[4] to curation at the 2025-2026 UABB biennale in Shenzhen and Hong Kong. Beneath the art runs this obsession with *research at scale*, sustained through four years organizing the program, another lineage that began at the Media Lab, in 2013.

Few settings other than the Media Lab enable a body of work to hold electronics, materials, textiles, interaction, manufacturing, and art together. This thesis needs all of them.

The fabric of this thesis is woven from all of them.

[1] https://scalablehci.com and https://ras.media.mit.edu
[2] Examples: https://era.world and https://bergsonne.io/
[3] https://media.mit.edu/posts/topographie-digitale-ars-electronica/
[4] https://media.mit.edu/publications/stymphalian-birds/

5 Timeline

These plans will depend on various uncertainties in administrative and technical aspects. The following timeline estimate might need to be compressed down to 1 year, in which case, some of the planned work could move to a postdoc. The final schedule and milestones will be determined in coordination with the committee.

<table>
<tr><th colspan="4">...PhD year 5 (2026)</th><th colspan="12">PhD year 6 (2027)</th><th colspan="5">PhD year 6.5 (2028)</th></tr>
<tr><th colspan="4">Fall</th><th>IAP</th><th colspan="4">Spring</th><th colspan="3">Summer</th><th colspan="4">Fall</th><th>IAP</th><th colspan="4">Spring</th></tr>
<tr><th>Sep</th><th>Oct</th><th>Nov</th><th>Dec</th><th>Jan</th><th>Feb</th><th>Mar</th><th>Apr</th><th>May</th><th>Jun</th><th>Jul</th><th>Aug</th><th>Sep</th><th>Oct</th><th>Nov</th><th>Dec</th><th>Jan</th><th>Feb</th><th>Mar</th><th>Apr</th><th>May</th></tr>
<tr><td>France visa</td><td>China manuf.</td><td colspan="6">Meta internship</td><td>China nanuf.</td><td colspan="3">Miniature sensing prototyping + manuf.</td><td colspan="4">Miniature electrostimulation prototyping + manufacturing</td><td>China nanuf.</td><td colspan="4">Dissertation + defense</td></tr>
<tr><td colspan="4">Cornell collaboration</td><td colspan="4">To confirm: Cornell collab.</td><td colspan="6">To confirm: EPFL + Poli Di Milano visitors</td><td colspan="7"></td></tr>
<tr><td colspan="4">To confirm: UROP</td><td colspan="5">To confirm: UROP</td><td colspan="3">To confirm: SFU visit.</td><td colspan="9"></td></tr>
</table>

Figure 5.1: Estimation of next steps, including collaborations with [visiting] students who have also been instrumental to achieve these explorations.

6 Personal Contributions

Main Publications

- Cedric Honnet, Wedyan Babatain, Yiyue Luo, Ozgun Kilic Afsar, Chloe Bensahel, Sarah Nicita, Yunyi Zhu, Andreea Danielescu, Neil Gershenfeld, and Joseph A. Paradiso. "FiberCircuits: A Miniaturization Framework To Manufacture Fibers That Embed Integrated Circuits." *ACM UIST*, 2025. DOI: https://doi.org/10.1145/3746059.3747802

- Cedric Honnet, Rachel Freire, Juliana Cherston, Maximilian Guenther, Joseph A. Paradiso, and Irmandy Wicaksono. "Ferrozuit: Ferromagnetic Electronic Textile System for Zero-Gravity Spatial Anchoring." *ACM UbiComp/ISWC Companion*, 2025. DOI: https://doi.org/10.1145/3714394.3750705

- Yunyi Zhu, Cedric Honnet, Yixiao Kang, Junyi Zhu, Angelina J. Zheng, Kyle Heinz, Grace Tang, Luca Musk, Michael Wessely, and Stefanie Mueller. "PortaChrome: A Portable Contact Light Source for Integrated Re-Programmable Multi-Color Textures." *ACM UIST*, 2024. DOI: https://doi.org/10.1145/3654777.3676458

- Cedric Honnet, Yunyi Zhu, Martin Nisser, Chao Liu, Byungchul Kim, Jae Hun Seol, Jongho Lee, Daniela Rus, and Stefanie Mueller. "RoboTouch: Laser-Etching Flexible Sensors for Robotic Touch Recognition." *IEEE ICRA*, 2023. Poster: http://doi.org/10.5281/zenodo.21879614

- Jun Gong, Olivia Seow, Cedric Honnet, Jack Forman, and Stefanie Mueller. "MetaSense: Integrating Sensing Capabilities into Mechanical Metamaterial." *ACM UIST*, 2021. DOI: https://doi.org/10.1145/3472749.3474806

- Thomas Preindl, Cedric Honnet, Andreas Pointner, Roland Aigner, Joseph A. Paradiso, and Michael Haller. "Sonoflex: Embroidered Speakers Without Permanent Magnets." *ACM UIST*, 2020. DOI: https://doi.org/10.1145/3379337.3415888

- Cedric Honnet, Hannah Perner-Wilson, Marc Teyssier, Bruno Fruchard, Jürgen Steimle, Ana C. Baptista, and Paul Strohmeier. "PolySense: Augmenting Textiles with Electrical Functionality using In-Situ Polymerization." *ACM CHI*, 2020. DOI: https://doi.org/10.1145/3313831.3376841

- Cedric Honnet and Gonçalo Lopes. "HiveTracker: 3D Positioning for Ubiquitous Embedded Systems." *ACM UbiComp/ISWC Adjunct*, 2019. DOI: https://doi.org/10.1145/3341162.3349295

Secondary Publications and Patents

- Rory S. Clark, Steve Hodges, Hyungyoung Kim, Valkyrie Savage, Oliver Child, Cedric Honnet, Joseph A. Paradiso, and Mike Fraser. "Are We Doing Enough? Sharing Design and Manufacturing Expertise Across Open Source Hardware Communities." *Extended Abstracts of the ACM CHI Conference on Human Factors in Computing Systems*, 2026. DOI: https://doi.org/10.1145/3772363.3778730

- Wedyan Babatain, Christine Park, Deiaa M. Harraz, Ozgun Kilic Afsar, Cedric Honnet, Sarah Lov, Jean-Baptiste Labrune, Michael D. Dickey, and Hiroshi Ishii. "Programmable Continuous Electrowetting of Liquid Metal for Reconfigurable Electronics." *Advanced Materials*, vol. 38, no. 2, e06383, 2026. DOI: https://doi.org/10.1002/adma.202506383

- Tianhong Catherine Yu, Cedric Honnet, Tingyu Cheng, Ryo Takahashi, Bo Zhou, Cheng Zhang, Paul Lukowicz, Yoshihiro Kawahara, Josiah Hester, Joseph A. Paradiso, Yiyue Luo, and Irmandy Wicaksono. "Intelligent Soft Wearables." *ACM UbiComp/ISWC Companion*, 2025. DOI: https://doi.org/10.1145/3714394.3750561

- Jack Forman, Ozgun Kilic Afsar, Sarah Nicita, Rosalie Hsin-Ju Lin, Cedric Honnet, Neil Gershenfeld, and Hiroshi Ishii. "Robotic liquid crystal elastomers, apparatus, and method of manufacture." U.S. Patent Application US20250136869A1, Application No. US18/927,122, published 2025, assigned to Massachusetts Institute of Technology. https://patents.google.com/patent/US20250136869A1/en

- Cedric Honnet, Tianhong Catherine Yu, Irmandy Wicaksono, Tingyu Cheng, Andreea Danielescu, Cheng Zhang, Stefanie Mueller, Joe Paradiso, and Yiyue Luo. "Democratizing Intelligent Soft Wearables." *Adjunct Proceedings of the ACM UIST Symposium*, 2024. DOI: https://doi.org/10.1145/3672539.3686707

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- Yiyue Luo, Junyi Zhu, Kui Wu, Cedric Honnet, Stefanie Mueller, and Wojciech Matusik. "MagKnitic: Machine-knitted Passive and Interactive Haptics Textiles with Integrated Binary Sensing." *ACM UIST*, 2023. DOI: https://doi.org/10.1145/3586183.3606765

- Olivia Seow, Cedric Honnet, Simon Perrault, and Hiroshi Ishii. "Pudica: A Framework For Designing Augmented Human-Flora Interaction." *ACM Augmented Humans Conference*, 2022. DOI: https://doi.org/10.1145/3519391.3519394

- Ozgun Kilic Afsar, Hila Mor, Cedric Honnet, and Hiroshi Ishii. "Choreographic Interfaces: Wearable Approaches to Movement Learning in Creative Processes." *ACM CHI 2021 Workshop on Human Augmentation for Skill Acquisition and Skill Transfer*, 2021. https://media.mit.edu/publications/choreographic-interfaces/

- Syamantak Payra, Irmandy Wicaksono, Juliana Cherston, Cedric Honnet, Valentina Sumini, and Joseph A. Paradiso. "Feeling Through Spacesuits: Application of Space-Resilient E-Textiles to Enable Haptic Feedback on Pressurized Extravehicular Suits." *IEEE Aerospace Conference*, 2021. DOI: https://doi.org/10.1109/AERO50100.2021.9438515

- Audrey Briot, Martin De Bie, Alice Giordani, Leon Denise, and Cedric Honnet. "Topographie Digitale." *ACM Tangible, Embedded, and Embodied Interaction (TEI)*, 2021. DOI: https://doi.org/10.1145/3430524.3444641

- Audrey Briot, Cedric Honnet, and Paul Strohmeier. "Stymphalian Birds – Exploring the Aesthetics of A Hybrid Textile." *Companion Publication of the ACM Designing Interactive Systems Conference*, 2020. DOI: https://doi.org/10.1145/3393914.3395840

- Paul Strohmeier, Narges Pourjafarian, Marion Koelle, Cedric Honnet, Bruno Fruchard, and Jürgen Steimle. "Sketching On-Body Interactions using Piezo-Resistive Kinesiology Tape." *ACM Augmented Humans Conference*, 2020. DOI: https://doi.org/10.1145/3384657.3384774

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Thesis and Book Chapter Contributions

- Justice Vidal. "Scalable Embedded Tiny Machine Learning (SETML): A General Framework for Embedded Distributed Inference." M.Eng. thesis, Massachusetts Institute of Technology, Department of Electrical Engineering and Computer Science, 2025. Cedric Honnet contributed project guidance. https://dspace.mit.edu/entities/publication/5eea2fe3-375e-47ed-a2cf-3b8b266fe132

- Kate Hartman. *Make: Wearable Electronics, Second Edition*. Make: Community, 2025. PolySense by Cedric Honnet et al. is featured in Chapter 7, "Sensors." https://www.oreilly.com/library/view/make-wearable-electronics/9781680457193/

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